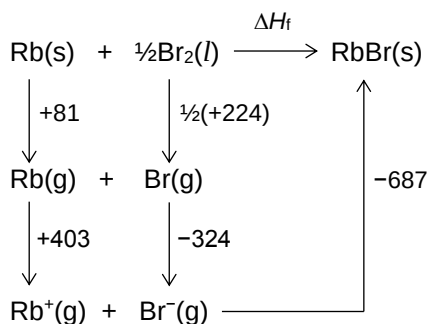


10



$$\begin{aligned}
 \Delta H_f &= +81 + \frac{1}{2}(+224) + 403 + (-324) + (-687) \\
 &= -415 \text{ kJ mol}^{-1}
 \end{aligned}$$

Answer: B

11 Heat evolved = (300 x c x ΔT) J

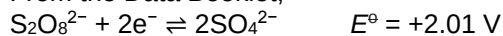
$$\text{No. of moles of propan-1-ol} = (m / 60.0) \text{ mol}$$

$$\begin{aligned}
 \text{Maximum amount of heat evolved} \\
 &= (2021 \times 1000 \times m / 60.0) \text{ J}
 \end{aligned}$$

$$\begin{aligned}
 \text{Efficiency of heating process} \\
 &= \frac{300 \times c \times \Delta T \times 60.0}{m \times 2021 \times 1000} \times 100\%
 \end{aligned}$$

Ans: D

12 From the Data Booklet,



In order for the metal ion to catalyse the reaction, $E^\ominus_{\text{cell}}$ must be positive. Hence the $E^\ominus$ value should be between +0.54 V and +2.01 V.

1: Incorrect. $E^\ominus(\text{Cu}^{2+}/\text{Cu}^+) = +0.15 \text{ V}$ 2: Correct. $E^\ominus(\text{Co}^{3+}/\text{Co}^{2+}) = +1.89 \text{ V}$ 3: Correct. $E^\ominus(\text{Mn}^{3+}/\text{Mn}^{2+}) = +1.54 \text{ V}$ 4: Incorrect. $E^\ominus(\text{V}^{3+}/\text{V}^{2+}) = -0.26 \text{ V}$

Ans: B

13 The rate of the reaction depends on the rate of the slow step.

$$\Rightarrow \text{rate} = k [\text{B}] [\text{C}]$$

However, the rate equation for the overall reaction should be in terms of concentrations of only the reactants (note that C and D are reaction intermediates).

Since step 1 of the mechanism is an equilibrium, $2\text{A} \rightleftharpoons \text{C}$, we can write an expression for the equilibrium constant:

$$K = \frac{[\text{C}]}{[\text{A}]^2}$$

rearranging, we get: $[\text{C}] = K [\text{A}]^2$

The rate equation then becomes

$$\begin{aligned}
 \text{Rate} &= k [\text{B}] K [\text{A}]^2 \\
 &= k' [\text{A}]^2 [\text{B}]
 \end{aligned}$$

Ans: C

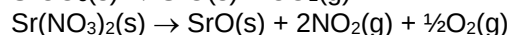
14. Option A suggest that element J is Al. Since in the aqueous solution Al^{3+} first form a white ppt with OH^- : $\text{Al}(\text{OH})_3$ which dissolves in excess NaOH due to the formation of $\text{Al}(\text{OH})_4^-$

Options B and D agrees with the suggestion that element J is Al.

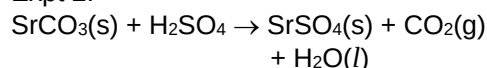
Option C is incorrect as Si has the highest melting point among the period 3 elements.

Answer: C

15 Expt 1:



Expt 2:



$$\text{No. of moles of CO}_2(\text{g}) = \frac{75}{V_m} \text{ mol}$$

(where V_m = molar volume at a certain temperature and pressure)

Let amount of SrCO_3 be a mol and amount of $\text{Sr}(\text{NO}_3)_2$ be b mol.

$$\text{From expt 1, } a + 2.5b = 200/V_m \text{ ----- eqn 1}$$

$$\text{From expt 2, } a = 75/V_m$$

Subst a = $75/V_m$ into eqn 1 gives

$$b = 50/V_m \text{ mol}$$

$$\begin{aligned}
 \Rightarrow \text{Mole ratio of SrCO}_3 : \text{Sr}(\text{NO}_3)_2 \\
 = 75/V_m : 50/V_m = 3 : 2
 \end{aligned}$$

Ans: A

16

	$E^\ominus$
$\text{Cl}_2 + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	+1.36
$\text{VO}_3^- + 4\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + 2\text{H}_2\text{O}$	+1.00
$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{VO}^{2+} + \text{H}_2\text{O}$	+1.00
$\text{VO}_2^+ + 2\text{H}^+ + \text{e}^- \rightleftharpoons \text{V}^{3+} + \text{H}_2\text{O}$	+0.34

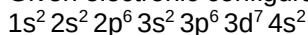
Cl_2 can only be reduced and VO^{2+} can only be oxidised.

Hence, VO^{2+} can be oxidised to either VO_3^- or VO_2^+ since the $E^\ominus$ values of $E^\ominus(\text{Cl}_2/\text{Cl}^-)$ is more positive than $E^\ominus(\text{VO}_2^+/\text{VO}^{2+})$ and $E^\ominus(\text{VO}_3^-/\text{VO}^{2+})$.

Answer: D

- 17 Highest oxidation state = number of 4s electrons + number of unpaired 3d electrons**

Given electronic configuration to be:



There are a total of two 4s electrons and three unpaired 3d electrons. Hence maximum oxidation state is +5.

O.S of Y in compound 1 is +6.

O.S of Y in compound 2 is +5.

O.S of Y in compound 1 is +4.

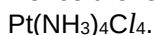
O.S of Y in compound 1 is +2.

Ans: A

- 18** Since the product is a platinum(IV) compound which contains a cation with a 2+ charge and has a co-ordination number of 6, this means that the cation contains two Cl^- ligands and four NH_3 ligands, i.e. $[Pt(NH_3)_4Cl_2]^{2+}$.

Since the reaction is between $PtCl_4$ and NH_3 only, and so the anion of the product is Cl^- ions, i.e. the product is made up of $[Pt(NH_3)_4Cl_2]^{2+}$ & $2Cl^-$.

Hence the formula of the product is:



Ans: C

- 19** The shortest C-H bonds are associated with those carbon hybrid orbitals with the greatest s character. With greater s character, the hybrid orbital can pull the electron density closer to the nucleus and results in shorter bond length.

A, B and **D** are correct statements but do not explain the C-H bond length.

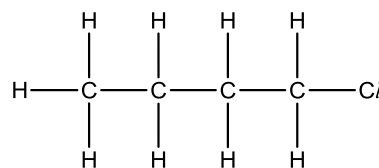
Ans: C

- 20** Chloroethane undergoes the S_N2 mechanism to give ethanol as it is a primary halogenoalkane. The mechanism involves an OH^- nucleophilic attack from the opposite end of the Cl leaving group and it is an overall second order reaction.

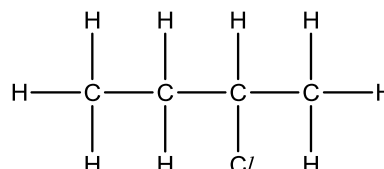
1, 3 and 4 are statements which describes S_N1 mechanism.

Ans: B

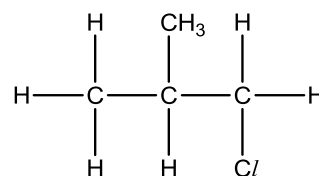
- 21** In total, there are 4 constitutional isomers that C_4H_9Cl can form. The isomers are as follows:



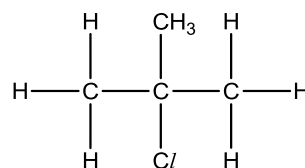
1-chlorobutane



2-chlorobutane



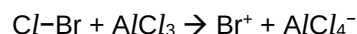
1-chloro-2-methylpropane



2-chloro-2-methylpropane

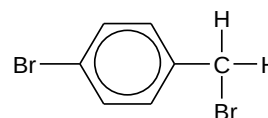
Ans: B

- 22** Since Br is less electronegative than Cl , the δ^+ charge will reside on the Br-atom in the $Cl-Br$ molecule. The molecule will therefore react with $AlCl_3$ to produce the electrophile Br^+ :



Br^+ then acts as the electrophile to attack the benzene ring.

Structure of G is therefore:



Ans: A

- 23** D undergoes dehydration to give alkenes E and F. E being the minor product as it is the less substituted alkene (i.e. less stable). Upon oxidation, E gives a ketone with $-\text{COCH}_3$ group.

A: Minor product obtained upon dehydration is $\text{CH}_2=\text{CHCH}_2\text{CH}_2\text{CH}_3$. It is oxidised to give CO_2 , H_2O and $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$.

B: The only product obtained upon dehydration is $\text{CH}_3\text{CH}=\text{CHCH}_2\text{CH}_3$. It is oxidised to give $\text{CH}_3\text{CO}_2\text{H}$ and $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$.

C: Minor product obtained upon dehydration is $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}=\text{CH}_2$. It is oxidised to give $\text{CH}_3\text{CH}(\text{CH}_3)\text{CO}_2\text{H}$, CO_2 and H_2O .

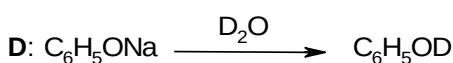
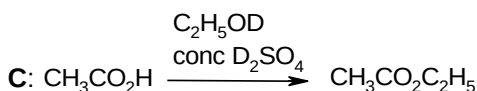
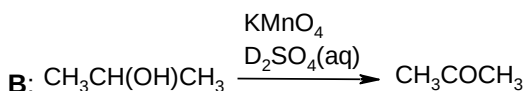
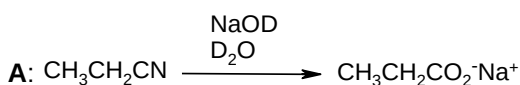
D: Minor product obtained upon dehydration is $\text{CH}_2=\text{C}(\text{CH}_3)\text{CH}_2\text{CH}_3$. It is oxidised to give CO_2 , H_2O and $\text{CH}_3\text{COCH}_2\text{CH}_3$.
 $\text{CH}_3\text{COCH}_2\text{CH}_3$ gives yellow ppt with I_2/OH^-

Ans: D

- 24** **A:** Both ethanol ($-\text{CH}(\text{OH})\text{CH}_3$) and propanone ($-\text{COCH}_3$) give yellow ppt with aqueous alkaline iodine.
- B:** Ethanol will be oxidised to ethanoic acid and turns acidified potassium dichromate(VI) from orange to green. Propanone cannot be oxidised and so acidified potassium dichromate (VI) will remain orange.
- C:** Propanone will give a yellow or orange ppt with 2,4 DNPH and ethanol will not give a ppt with 2,4 DNPH.
- D:** Effervescence of H_2 gas which 'pops' with lighted splint will be observed for ethanol reacting with sodium metal. No gas will be evolved when sodium metal is added to propanone.

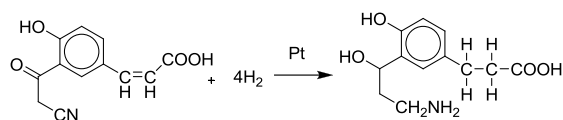
Ans: A

25



Ans: D

26 **1: Correct**



1 mole of compound **P** reacts completely with 4 moles of H_2 gas.

2: Correct

It has 1 $\text{C}=\text{C}$ which can exhibit cis-trans isomerism and no chiral carbon.

Total number of stereoisomers = $2^1 = 2$

3: Incorrect

Phenol does not react with ethanoic acid in the presence of concentrated sulfuric acid. Acid chloride is needed to react with phenol to form ester.

Ans: C

- 27** **N_D** is the most basic as it has an electron donating methyl group on the adjacent carbon which makes the lone pair of electrons more available for donation to a proton. Hence, **N_D** is more basic than **N_C**.

Lone pair of electrons on **N_B** is delocalised into the pyridine rings (similar to phenylamine) which makes the lone pair of electron less available for donation to a proton. Hence, **N_B** is less basic than **N_C** and **N_D** where the lone pair of electrons are not delocalised into the pyridine rings.

N_A is an amide nitrogen with the lone pair of electron being delocalised into the adjacent carbonyl group. Hence the lone pair is not available for donation to a proton. Amide is neutral.

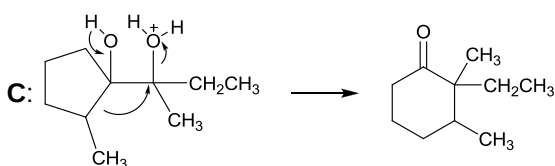
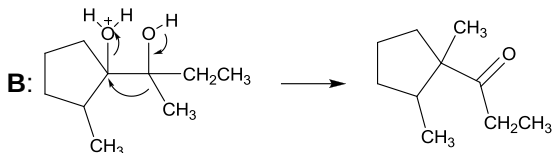
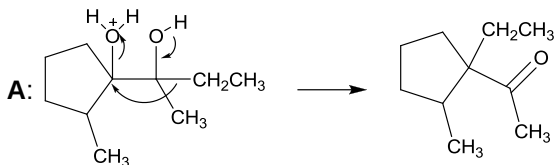
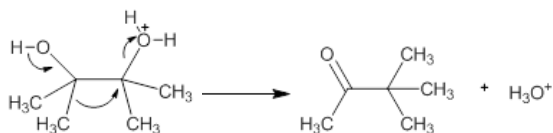
Order of basicity is $\text{N}_\text{D} > \text{N}_\text{C} > \text{N}_\text{B} > \text{N}_\text{A}$

Ans: D

- 28** **A:** Correct. 2-aminopropanoic acid isolated from silkworm has a chiral centre and is optically pure and is able to rotate plane polarised light.
- B:** Incorrect. Amine cannot react with ethanoic acid to form an amide. Acyl chloride which is more reactive has to be used to react with amine.
- C:** Incorrect. An aqueous solution of 2-aminopropanoic acid exists as zwitterions and has buffering capacity.
- D:** Incorrect. Aqueous bromine is decolourised only when alkene, phenol or phenylamine is present (and is absent in 2-aminopropanoic acid)

Ans: A

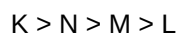
- 29** The pinacol rearrangement works via the mechanism as shown.



Compound **A**, **B** and **C** can be obtained via pinacol rearrangement.

Ans: D

- 30** Rate of hydrolysis is in the following order:



K is acid chloride and undergoes hydrolysis most readily due to the presence of 2 electronegative atoms (Cl and O atoms) attached to the carbon.

For halogenoalkane, rate of hydrolysis depends on the strength of the C-Hal bond. C-Br bond is weaker than C-Cl bond and hence is broken more readily.

L is a chlorobenzene and the C-Cl bond is not broken in the presence of NaOH(aq). The lone pair of electrons on Cl is partially delocalised into the benzene ring, leading to partial double character and strengthen the C-Cl bond.

71.1 g of AgCl is equivalent to 0.5 mol of AgCl. W will give more AgCl as it is more reactive than M.

N will give more than 0.5 mol of AgBr which is equivalent to 93.9 g of AgBr.

L will not give any precipitate.

Ans: A